

Multitap Comb Reverb — Design Notes

Context for collaborating agents: designing a 2D-navigated multitap comb intended for reverberation. User position moves through a field of taps; distance drives gain, with delay length ("size"), pan, and gain all smoothed.

Problem 1: Impulse/click artifacts when user passes near a tap

Symptom: Click or burst-like impulses when the navigation control passes close to a tap, despite gain, size, and pan smoothing being in place.

Root cause: Smoothing bounds parameter *values* but not *rates of change*. Near a tap, $dParam/dt$ explodes relative to user motion. Singularities in the math survive smoothing.

Likely culprits (in order)

1. **Delay-time modulation (if "size" = delay length).** Near a tap, the read head can move >1 sample per output sample, which clicks under linear/cubic interpolation. Smoothing the parameter trades the click for a chirp.
 - **Fix:** Dual read heads with crossfading when delay changes ("Doppler-free" / crossfading delay). Allpass interpolation helps but is not a complete solution.
2. **Pan/azimuth singularity.** atan2 to a tap flips $\sim 180^\circ$ as you cross through it. A scalar smoother on the angle either lags heavily or pops.
 - **Fix:** Smooth pan as a 2D vector (x/y or sin/cos pair) and re-derive angle. Or fade spatialization weight by $1/(r+\epsilon)$ so orientation stops mattering near zero.
3. **Gain cap is a C^0 corner.** $\text{clamp}(k/r, 0, \text{max})$ is continuous but not differentiable.
 - **Fix:** Soft cap, e.g. $\text{max} \cdot \tanh(k/(r \cdot \text{max}))$, keeps the derivative continuous.

Diagnostic

Log per-sample delta of each tap's delay length and pan angle during the artifact. Whichever spikes is the culprit.

Pragmatic fix (often resolves all three at once)

Enforce a minimum radius $r \geq \varepsilon$ (e.g. 5% of field radius) so the user can't physically pass through a tap. Removes the singularities; reverbs aren't supposed to let you stand inside a wall reflection anyway.

Problem 2: Sounds recognizably like a multitap comb, want more interest

Why it sounds like a comb: All taps are identical-character copies summed at integer-ish delay relationships, producing predictable spectral peaks. To break the sound, break (a) similarity of taps, (b) linearity of the system, or (c) stationarity of the tap structure.

Highest-leverage moves (turn delay into reverb)

- **Per-tap pitch shift**, even ± 5 –30 cents randomized per tap. Destroys harmonic reinforcement that creates comb peaks. Larger shifts (fifths, octaves) $\rightarrow$ shimmer.
- **Per-tap modulated allpass diffusion**. Short chain of allpasses per tap with slow random LFO on the coefficient (Schroeder/Dattorro trick). Smears each tap's transient so taps stop sounding like discrete echoes.
- **Independent delay-time modulation per tap**, different LFO rates/phases. Even a few samples of wander breaks comb resonance into something choral.

Per-tap character (stacking random processing)

- Random per-tap filtering — bandpass with random center, lowpass with random cutoff, comb-within-comb. Filters can drift so timbre evolves.
- Waveshaping/saturation per tap (different curves per tap). Nonlinearity introduces inharmonic content the comb can't predict.
- Bitcrush / SR-reduce a subset of taps, especially when modulated.
- Ring-mod a tap against a slow sine or against another tap.
- Reverse-buffer reads on some taps.

Structural / topological

- **Cross-feedback between taps**, not just to the sum. Essentially a sparse FDN; with a unitary (Hadamard or Householder) feedback matrix this yields proper diffuse reverb tails from the same skeleton.
- **Tap respawn**: taps have lifetimes, fade out and reappear at new random positions. Tail

evolves over time.

- **Tap probability** as a function of position — some regions dense, others sparse.

Glitch / discontinuous controls

- **Sample-and-hold on tap positions** at irregular trigger rates — quantum-jump topology. Smoothing OFF on these is the point.
- **Per-tap stutter/freeze:** tap occasionally latches its current buffer contents and loops a slice for N ms.
- **Probabilistic tap mute/unmute** on a clock, like a Euclidean rhythm across the tap field.
- **Transient-triggered events:** input transient causes a random tap to jump position, freeze, or pitch-shift momentarily.
- **Buffer scrub:** read pointer for a tap occasionally jumps to a random position in the delay line.
- **Granulate at the tap output:** each tap emits grains rather than a continuous read, with randomized grain size/pitch/window.

Two design philosophies (pickable, or crossfade between)

1. **Continuous lushness** — pitch shift + allpass diffusion + delay modulation. Sounds like a reverb, no longer like a comb.
2. **Glitchy/event-driven** — S&H controls, probabilistic gates, transient-triggered jumps. Sounds alive and unpredictable, still recognizably a multitap.

Running both in parallel and crossfading via a macro gets a huge range out of one tap structure.